



# TEXT CLASSIFICATION PROJECT

GROUP 5 MEI  
APRENDIZAGEM PROFUNDA

By : Paul Brunner  
Francesco Scivoletto  
Moritz Zideck





# PROJECT GOALS

01

Identify whether a short English text is human-written or generated by an AI model, and classify it into the correct source family.

02

Compare traditional and deep learning approaches to find the most accurate and robust model for external validation and final testing.



# DATASET - MODELS

## SOURCE

Hello-SimpleAI/HC3

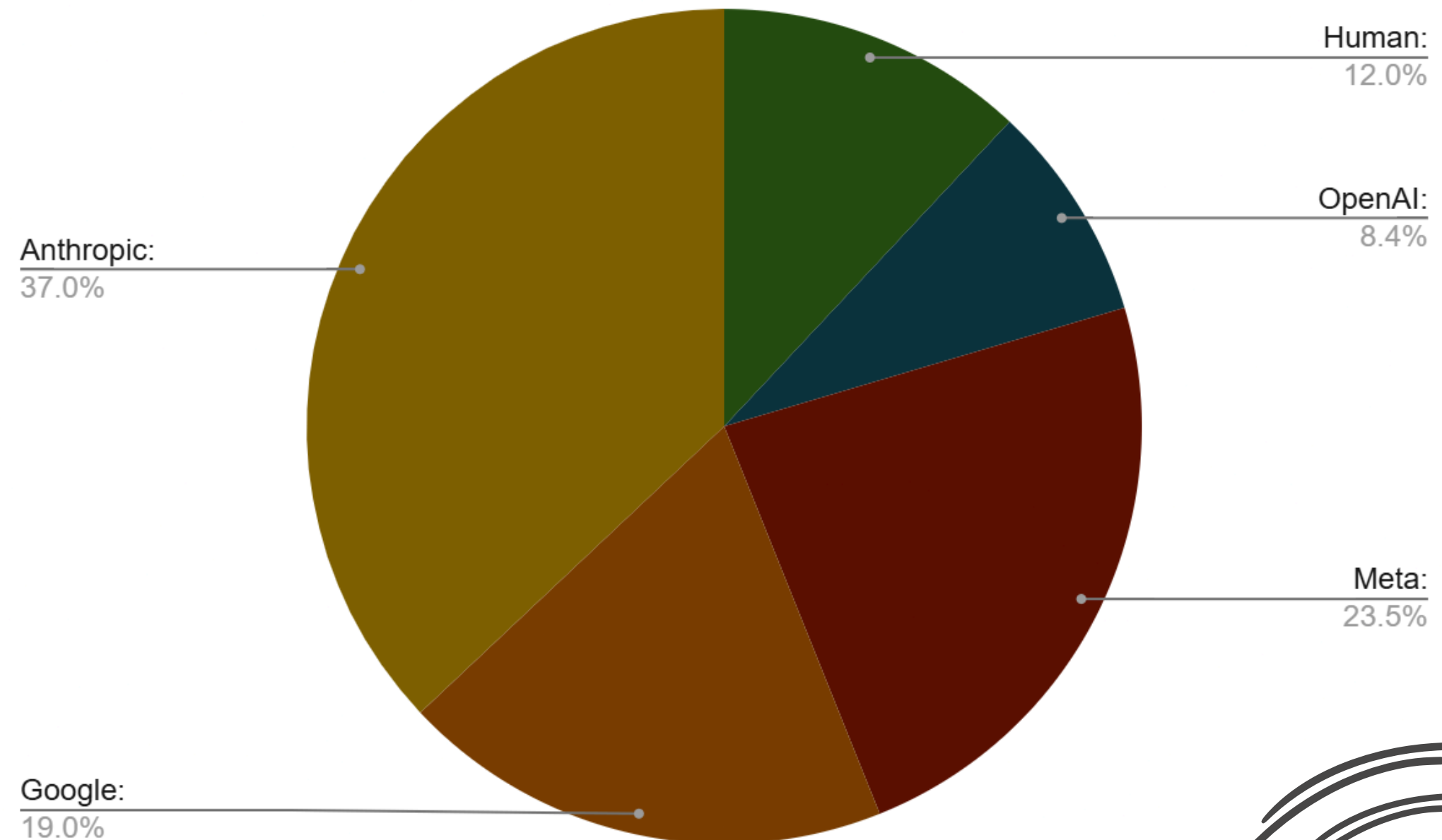
MLNTeam-Unical/OpenTuringBench

Norquinal/claude\_multiround\_chat\_30k

Different models (Companies) and their counts:

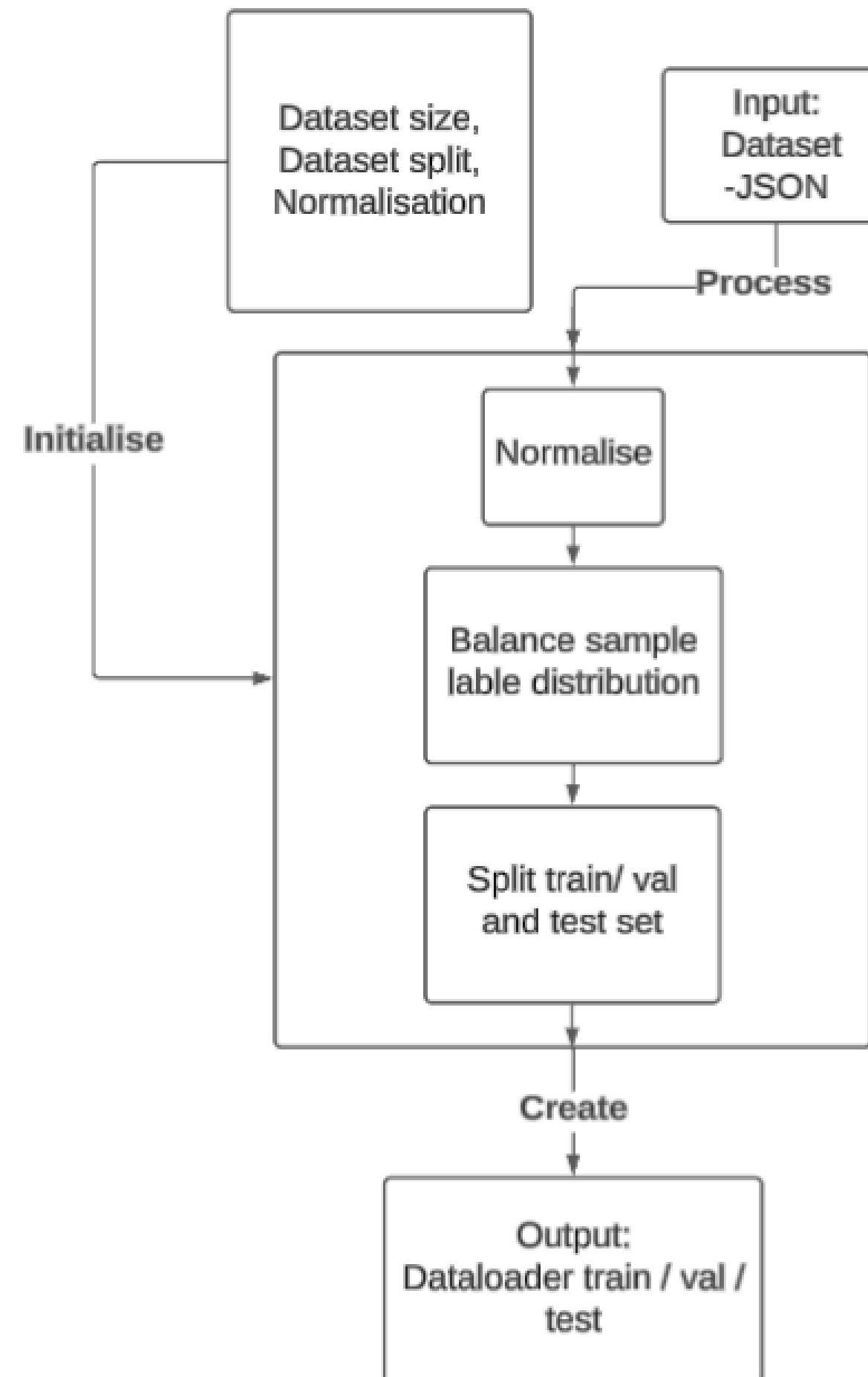
- Human: 33782
- OpenAI: 23788
- Meta: 66311
- Google: 53716
- Anthropic: 104440

Total samples: 282,037 | Classes: 5



# TEXT PREPROCESSING AND DATA LOADING

- Unicode **normalization**
- Whitespace and newline cleanup
- **Balanced** class sampling
- Deterministic shuffling
- Train / validation / test **split**
- Frequent word extraction
- **Configurable** dataset size

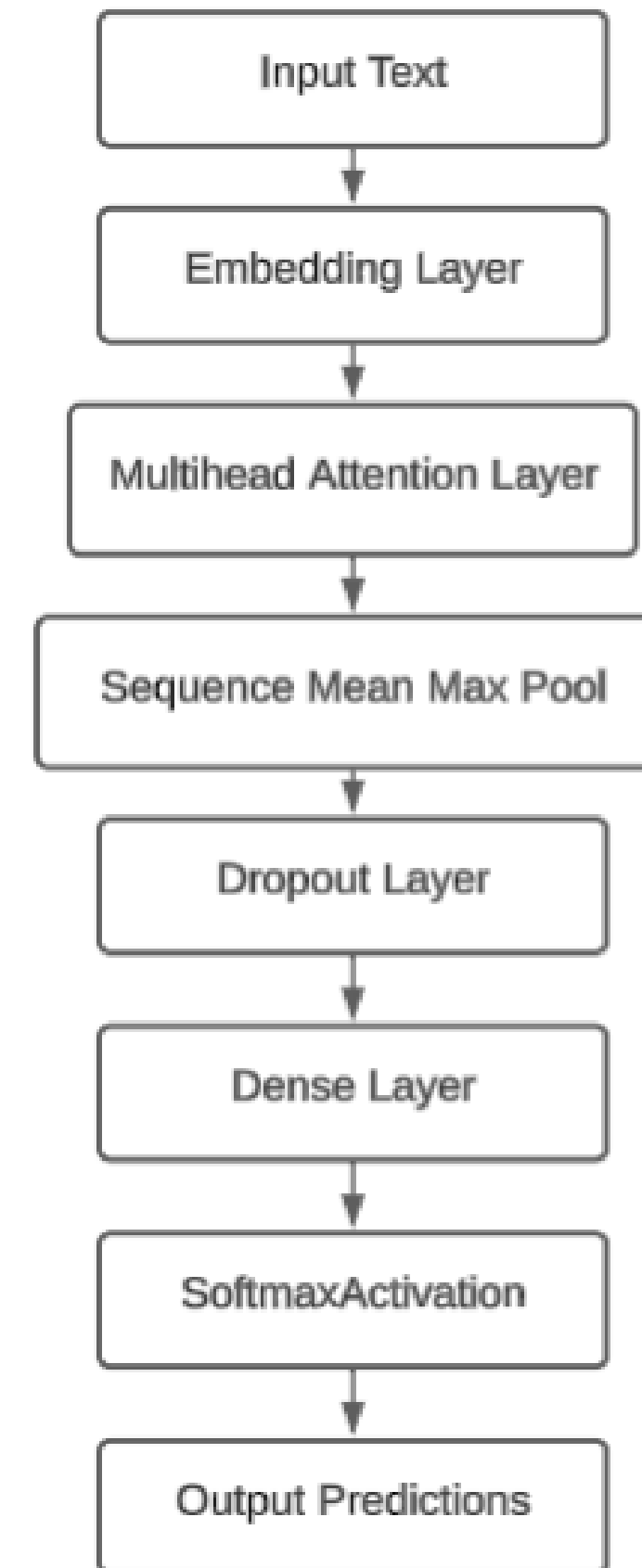


# DEEP NEURAL NETWORK WITH MULTIHEAD ATTENTION

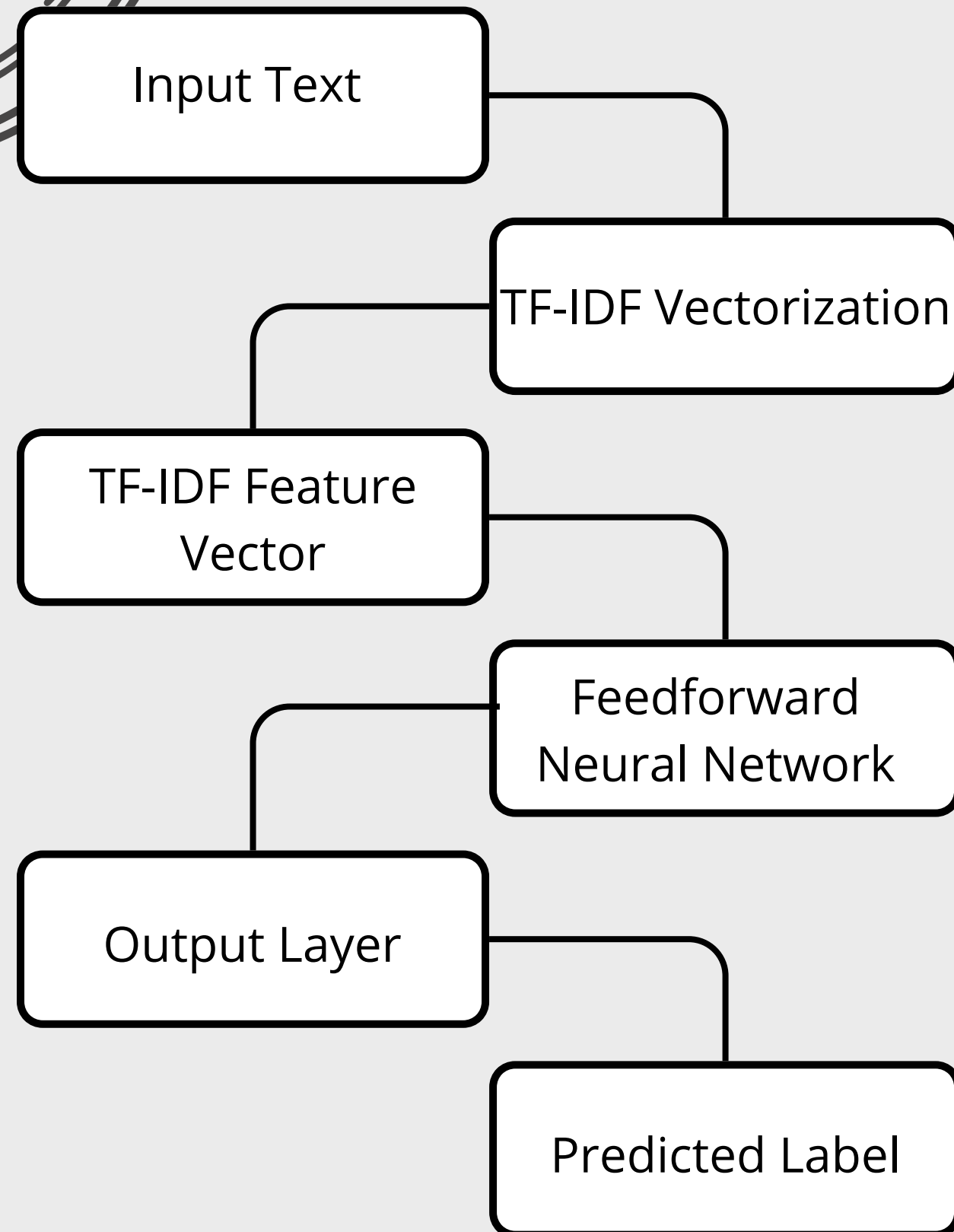
- Better at capturing long-range dependencies than GRU
- **Multi-Head Attention** learns different types of relationships in the sequence
- Can focus on **multiple relevant parts of the input** at the same time
- **Hyperparameters:**
  1. Embedding size: 128
  2. Vocabulary size / encoded words: 30,000
  3. Number of attention heads: 4
  4. Dropout: 0.3
  5. Epochs: 50

## RESULTS:

- Accuracy on training / own dataset: 0.82
- Accuracy on evaluation dataset: TBD

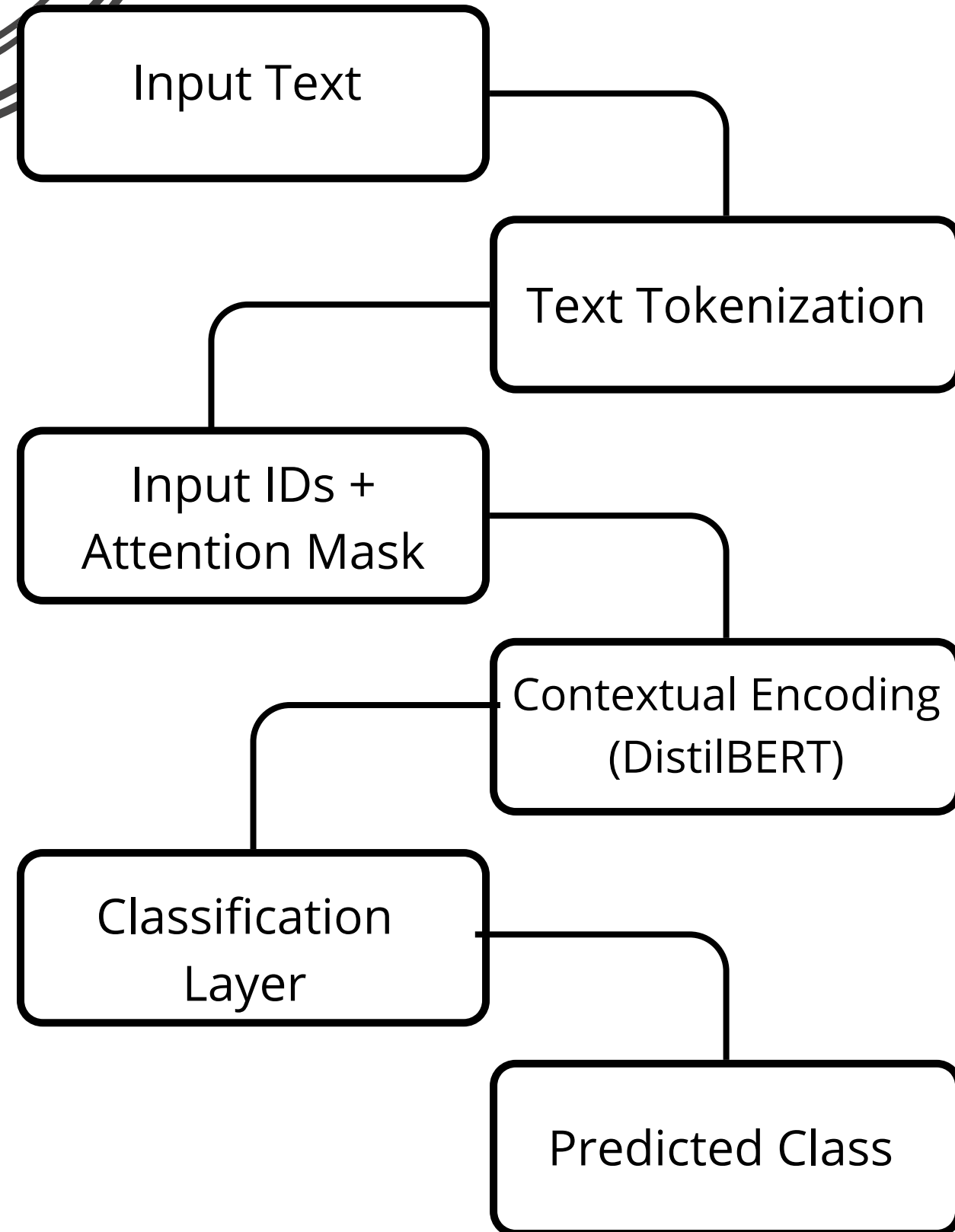


# TF-IDF + FEEDFORWARD NEURAL NETWORK



- **Model type:** classical text classification pipeline combining **TF-IDF feature extraction** with a **PyTorch feedforward neural network**.
- **Input representation:** each text is converted into a **5000-dimensional TF-IDF vector**, capturing the importance of words in the document.
- **Optimization pipeline:** data cleaning, label mapping, label encoding, **80/20 stratified split**, TF-IDF fitted on training data only, training with **CrossEntropyLoss** and **Adam** optimizer (lr = 0.001) for **5 epochs**.
- **Use in submissions:** the trained model, TF-IDF vectorizer, and label encoder were saved and reused to preprocess unseen texts, generate predictions, and produce the final submission labels.

# DISTILBERT TRANSFORMER FOR TEXT CLASSIFICATION



- **How it works:** the text is tokenized into subwords and processed by DistilBERT, which builds a contextual sentence representation. A classification head then predicts the most likely label.
- **Why it is different:** unlike TF-IDF models, it uses context and semantic information, not only word frequency.
- **Training pipeline:** cleaning, label encoding, stratified 80/20 split, tokenization (max length 256), fine-tuning with learning rate  $2e-5$ , batch size 16, 3 epochs, and early stopping.
- **Use in submissions:** the fine-tuned model predicts labels for unseen texts and produces the final submission file.

# Results



## TF-IDF + FEEDFORWARD NEURAL NETWORK

validation accuracy

**80,01%**

professor file acc

**38,4%**



## DISTILBERT TRANSFORMER

validation accuracy

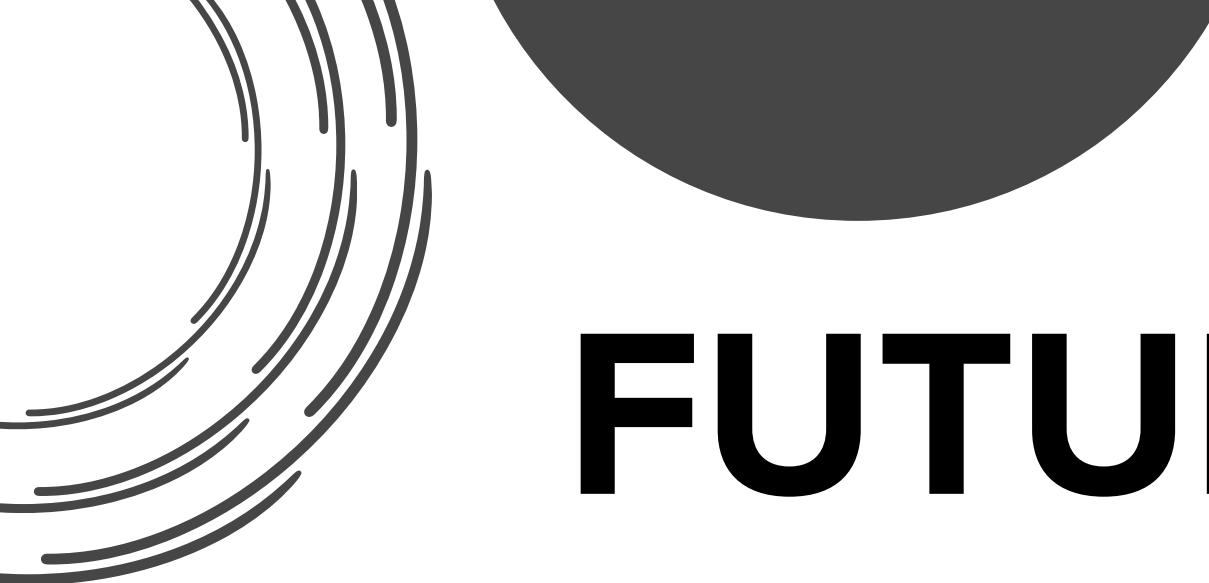
**92,33%**

professor file acc

**44,0%**

**Main takeaway:** DistillBERT outperformed the TF -IDF+FFNN baseline on both validation data (+12.33 points) and the external professor set(+5.6 pints)

Why this matters: Uses contextual representations, while the TF-IDF model relies on word-importance feature only.



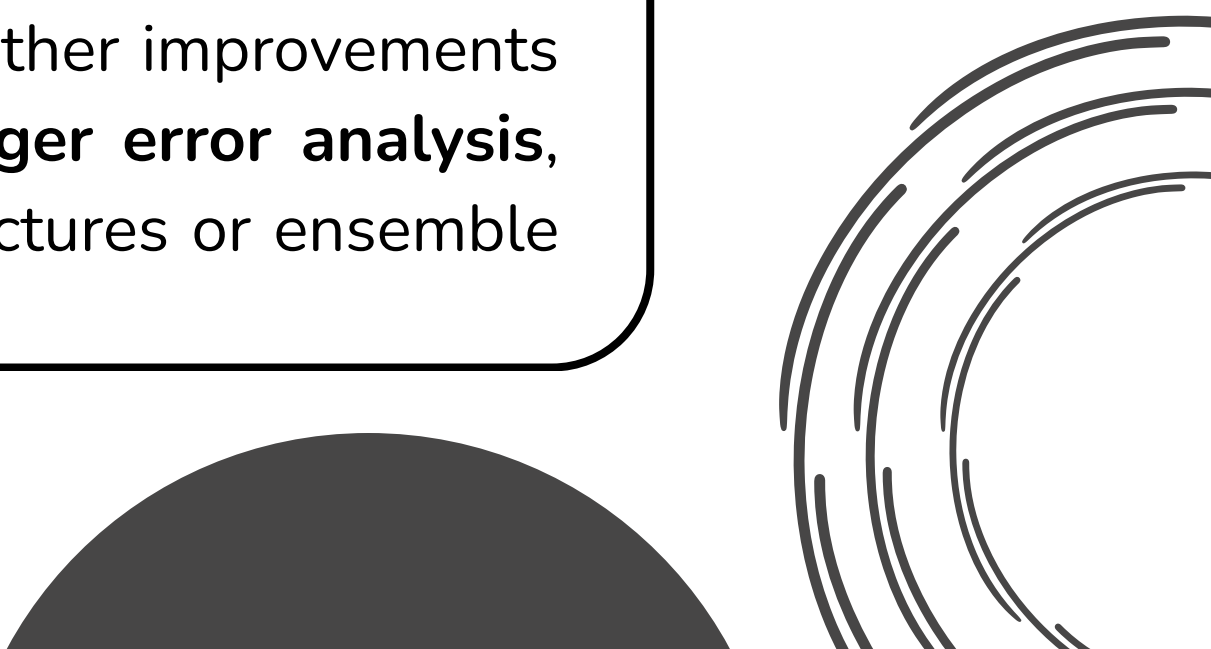
# FUTURE DEVELOPMENT

## DNN

Future improvements for the DNN model include a **deeper architecture** with more attention heads, **stronger embeddings** based on pretrained or subword representations, and more systematic **hyperparameter tuning** with longer training. Performance could also benefit from improved pooling strategies, **data augmentation**, and better class balancing

## BERT

For BERT, future work could focus on **hyperparameter optimization**, better class balancing, and **domain-adaptive fine-tuning** on texts closer to the target dataset. Further improvements may also come from data augmentation, **stronger error analysis**, and comparison with newer transformer architectures or ensemble approaches.





**THANKS FOR  
ATTENTION**